

AWS Resource Access Manager (RAM) - Lab Guide

Lab Objective

Set up and validate cross-account resource sharing using **AWS Resource Access Manager (RAM)**. In this lab, Account A will share a **VPC Subnet** with Account B.

Prerequisites

- Two AWS accounts:
 - **Account A** (Resource Owner)
 - **Account B** (Resource Consumer)
 - AWS Organizations (optional, but simplifies auto-accept)
 - Administrator IAM access in both accounts
-

Step 1: Create a VPC and Subnet in Account A

1. Login to **Account A** via AWS Console.
2. Navigate to **VPC > Your VPCs > Create VPC**:
 - **Name:** Shared-VPC
 - **CIDR block:** 10.0.0.0/16
3. Go to **Subnets > Create Subnet**:

- **VPC:** Shared-VPC
 - **Subnet CIDR:** 10.0.1.0/24
 - **Name:** Shared-Subnet
-

Step 2: Create a Resource Share in Account A

1. Navigate to **AWS RAM > Resource shares**.
2. Click **Create resource share**.
3. Enter:
 - **Name:** Share-VPC-Subnet
4. Under **Resources**:
 - Select **VPC Subnets**, and choose **Shared-Subnet**
5. Under **Principals**:
 - Choose **Allow sharing with AWS accounts**
 - Enter the **Account ID of Account B**
6. Review and **Create resource share**

☒ If Account A and B are in the same AWS Organization, enable "Allow sharing within your organization" and Account B will auto-accept the share.

Step 3: Accept the Share in Account B

1. Login to **Account B**

2. Navigate to **AWS RAM > Shared with me**
 3. Locate the share **Share-VPC-Subnet**
 4. Click **Accept resource share**
-

Step 4: Use Shared Subnet in Account B

1. Go to **EC2 > Launch Instance**
2. In **Network settings**:
 - Select **Shared-VPC** and **Shared-Subnet** (it appears automatically)
3. Complete instance launch

 Ensure IAM permissions in Account B allow use of shared subnets.

Step 5: Validation

- **In Account B:**
 - Verify the EC2 instance is using the subnet from Account A.
 - **In Account A:**
 - Use **VPC Flow Logs** or **CloudWatch Logs** to monitor network traffic.
-

Step 6: Cleanup

- **In Account B:**
 - Terminate the EC2 instance.

- In **Account A**:
 - Delete the Resource Share from AWS RAM.
 - Delete the Subnet and VPC.
-

Notes

- Resource sharing is secure and doesn't duplicate resources.
 - Billing remains with the **resource owner (Account A)**.
 - You can also share other resources like Transit Gateways, Route 53 Resolver rules, etc.
-

Additional Exercises

- Share a **Transit Gateway**.
 - Share **Route 53 Resolver Rules**.
 - Use **Tag-based policies** for fine-grained access control.
-

References

- AWS RAM Documentation:
<https://docs.aws.amazon.com/ram/latest/userguide/what-is.html>
- VPC Sharing Best Practices:
<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-sharing.html>